Mingyue Huo 霍明月 /mɪŋɥœ xuɔ/

mhuo5@illinois.edu| LinkedIn | GitHub | Google Scholar
Seeking Full-Time Research Scientist roles | Available Dec 2026

Education

University of Illinois Urbana-Champaign USA
PhD in Computational Linguistics *Dec 2026 (expected)*
MS in Applied Statistics *May 2026*
MA in Japanese Linguistics *May 2022*

Shanghai International Studies University China
BA in Japanese *June 2020*

Hosei University (exchange) *2018*, Tokyo, Japan

Highlights

Industry Research: Five research internships across Netflix, Tencent AI Lab, Amazon Prime Video, ByteDance AI Lab, and MERL, spanning speech, audio, and multimodal AI.

Research Focus: Speech and multimodal AI, with recent work focused on foundation-model evaluation and post-training, built on prior research in speech representation learning and multi-speaker understanding.

Publications: 7 first-author publications in leading speech and NLP venues, including ICASSP, Interspeech, ACL, ASRU, and the *Journal of the Acoustical Society of America*.

Open-source / Demos: end-to-end demos for Speaker Diarization (time-stamped meeting transcription for real-world application) and Target Speaker Extraction. Contributor to open-source audio understanding toolkit Auden.

Publications

SpeechCritic: Learning a Diagnostic Speech Judge from Limited Human Preferences
Mingyue Huo, Shivam Mehta, Bhavin Jawade, Yinghong Lan, Haoqi Li
In submission [PDF] [Demo]

Louder, Longer, Livelier: Acoustic Shortcuts and Misgrounded Rationales in Speech LLM Judges
Mingyue Huo, Shivam Mehta, Bhavin Jawade, Yinghong Lan, Haoqi Li
In submission [PDF] [Benchmark]

Where Speech Enhancement Hurts Recognition: An Inference Time Polar Projection Diagnosis
Mingyue Huo, Yuheng Zhang, Hao Zhang
In submission [PDF]

Fooling Reward Models with Gibberish: Adversarial Examples Beyond the Semantic Regime
Yuheng Zhang∗, **Mingyue Huo**∗, Minghao Zhu, Mengxue Zhang, Nan Jiang (∗ Equal contribution)
In submission [PDF]

TagSpeech: End-to-End Multi-Speaker ASR and Diarization with Fine-Grained Temporal Grounding
Mingyue Huo, Yiwen Shao, Yuheng Zhang
ACL 2026 Main **Oral** [PDF] [Code+Demo] [Model]

Revisiting Audio-language Pretraining for Learning General-purpose Audio Representation
Wei-Cheng Tseng, Xuanru Zhou, **Mingyue Huo**, Yiwen Shao, Hao Zhang, Dong Yu
ACL 2026 Main [PDF]

Auden-Voice: General-Purpose Voice Encoder for Speech and Language Understanding
Mingyue Huo, Wei-Cheng Tseng, Yiwen Shao, Hao Zhang, Dong Yu
ICASSP 2026 [PDF] [Code] [Model]

Identifying and Calibrating Overconfidence in Noisy Speech Recognition
Mingyue Huo, Yuheng Zhang, Yan Tang
ASRU 2025 [PDF]

Beyond Speaker Identity: Text Guided Target Speech Extraction
Mingyue Huo, Abhinav Jain, Cong Phuoc Huynh, Fanjie Kong, Pichao Wang, Zhu Liu, Vimal Bhat
ICASSP 2025 [PDF] [Demo]

Iterative Nash Policy Optimization: Aligning LLMs with General Preferences via No-Regret Learning
Yuheng Zhang, Dian Yu, Baolin Peng, Linfeng Song, Ye Tian, **Mingyue Huo**, Nan Jiang, Haitao Mi, Dong Yu
ICLR 2025 **Oral** [PDF]

What makes listening comprehension difficult?: A feature-based machine learning approach to understanding item difficulty
Huiying Cai, Xun Yan, Ping-Lin Chuang, Yulin Pan, **Mingyue Huo**
Journal: Applied Linguistics, 2025 [PDF]

Release from Same-talker Speech-on-speech Masking: Effects of Masker Intelligibility and Other Contributing Factors
Mingyue Huo, Yinglun Sun, Dan Fogerty, Yan Tang
Journal: Journal of the Acoustical Society of America (JASA), 2024 [PDF]

Quantifying Informational Masking due to Masker Intelligibility in Same-talker Speech-in-speech Perception
Mingyue Huo, Yinglun Sun, Dan Fogerty, Yan Tang
INTERSPEECH 2023 **Oral** [PDF]

Towards Automating the Assessment of Teachers' Teaching Emotions in Preschool Classrooms
Mingyue Huo, Yan Tang
20th International Congress of Phonetic Sciences (ICPhS) 2023 [PDF]

Industry Experience

Research Intern at MERL (Mitsubishi Electric Research Laboratories) Fall 2026, Cambridge, MA
Host: Jonathan Le Roux

- Long-form audio understanding in real-world multi-speaker meeting scenarios.

Research Scientist Intern at Netflix Summer 2026, Los Gatos, CA
Host: Yinghong Lan

- Developed a diagnostic Audio LLM-as-a-Judge for cross-lingual generation and systematically evaluated Supervised Fine-Tuning (SFT), On-Policy Distillation (OPD), and Reinforcement Learning (RL) as post-training strategies.
- Designed a supervision amplification framework that uses calibrated acoustic signals as soft privileged information for Gemini, scaling scarce human supervision into large training sets.

Research Scientist Intern at Tencent AI-Lab Summer 2025, Bellevue, WA
Host: Dong Yu

- Developed general-purpose speech representations by systematically adapting and benchmarking representation-learning methods from vision for speech, evaluating learned representations across speaker identity, paralinguistics, speech–text retrieval, and audio-language understanding.
- Built general-purpose speech encoders for speaker verification, diarization, speaker counting, and LLM-based audio QA; contributed to the open-source Auden toolkit, with resulting research accepted at ICASSP 2026.

Applied Scientist Intern at Amazon Prime Video Summer 2024, Seattle, WA
Host: Zhu Liu

- Developed a text-guided target speech extraction model that conditions on natural-language descriptions to identify and extract target speakers without traditional audio identifiers.

- Built a speech–mixture–text–description dataset and evaluation pipeline, achieving an average 16 dB SI-SDR improvement; resulting research was accepted at ICASSP 2025.

Linguist Intern at ByteDance AI-Lab Dec 2019 – July 2021, Shanghai, China

- Worked on Mandarin and Japanese TTS and singing synthesis, designing annotation pipelines, grapheme-to-phoneme conversion rules, and linguistically targeted evaluation sets for model development.
- Evaluated generative speech models using expert analysis, MOS, and A/B testing to improve naturalness and expressiveness; several resulting voices were launched on TikTok.

Awards

- UIUC Graduate College Dissertation Completion Fellowship for 2026-2027
- Conference Travel Award – UIUC Linguistics November 2025
- C.W. Kim Research Award in Linguistics – UIUC May 2024
- Linguistics Student Leadership Award – UIUC May 2024
- Midwest Speech and Language Days Travel Award – University of Michigan April 2024
- Departmental Conference Travel Award – UIUC November 2023
- Best Japanese TA – UIUC May 2022
- EALC Summer Research Fellowship – UIUC May 2021
- Teachers Ranked as Excellent by Their Students – UIUC 2021 Fall and 2020 Fall

Community Service

- **Student Volunteer** at INTERSPEECH 2021, NeurIPS 2023
- **Reviewer** at ICASSP 2026, INTERSPEECH 2025 & 2026, SLT2026

Skills

- **Language:** Mandarin (Native), English (Advanced), Japanese (Advanced N1)
- **Software:** Praat, Audacity, Adobe Audition, Adobe Premiere, Montreal Forced Aligner
- **Programming:** Python, TensorFlow, MATLAB, Shell, R, LaTeX
- **Framework & Library:** PyTorch, Kaldi, NumPy, SciPy, Torchaudio, librosa, soundfile, Matplotlib, Lhotse
- **Highlighted Coursework:** Digital Signal Processing (A+), Psychoacoustics (A), Phonetics (A+), Phonology (A), Formal Semantics (A+), Syntax (A), Machine Learning (A), Natural Language Processing (A), Generative Models for Audio (A)

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